CHANGE REQUEST

CR-05

Skills Wallet, Interactive Skill Tree & Gamification Module

Base SRS: Sulfri Trainer Portal — Personal Brand Edition v1.0
Amendment extends: CR-01 (Badge Wallet) + CR-02 (Skill Tree) + CR-03 (Event Lead Capture) + CR-04 (Showcase & Automation)

CR ID	CR-05
Type	Functional Enhancement — Credential Visualisation, Skill Intelligence & Engagement
Priority	High
Target Version	v1.5 (Post-Approval Consolidation)
Base SRS	Sulfri Trainer Portal — Personal Brand Edition v1.0
Extends	CR-01 (Badge Wallet) + CR-02 (Skill Tree foundation) + CR-03 + CR-04
Implementation	src/app/skills/page.tsx — production-ready component delivered
Prepared By	System Specification Draft
Status	Pending Approval — Implementation Code Available

Module A	Module B	Module C
Skills Wallet (Verified Credentials Grid)	**Interactive Radial Skill Tree (SVG)**	**Gamification Engine (XP, Levels, Achievements)**

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1. EXECUTIVE SUMMARY

This Change Request formalises and specifies the complete Skills Wallet, Interactive Radial Skill Tree, and Gamification Engine for the Sulfri Trainer Portal. While CR-02 introduced the concept of a skill tree and expertise linking, it did not specify the visual interaction model, gamification mechanics, or the precise mapping between the existing training class database and the skill/expertise structure.

CR-05 closes this gap by delivering: (1) a production-ready implementation reference for the Skills Wallet with Credly-verified badge integration; (2) a fully interactive radial SVG skill tree built without external graphing dependencies, derived dynamically from the live training class and badge database; and (3) a gamification layer — XP points, 5-level progression, and an achievement system — that rewards and communicates Sulfri's accumulated expertise in an engaging, verifiable format.

Module A	**Skills Wallet** Tabbed credential showcase: Credly badge grid with verification links, domain filter chips, skill area cards with badge thumbnails and training class counts.
Module B	**Interactive Radial Skill Tree** Pan-and-zoom SVG radial tree: Root (trainer) → Domain → Topic nodes, derived from live class database. Click-to-inspect panel shows supporting badges and training history per node.
Module C	**Gamification Engine** XP = (badges × 100) + (classes × 10). Five levels: Trainee through Master Trainer. Five unlockable achievements. Live XP progress bar with gradient glow. All computed client-side from existing API data.

2. CURRENT STATE ANALYSIS

2.1 What Exists (test/agent-kit branch + live deployment at sulfri-portal.vercel.app)

- Badge model (Prisma): id, title, slug, issuer, credlyBadgeId, iframeWidth, iframeHeight, fallbackImageUrl, verificationUrl, featured, visibility, displayOrder.
- Skill model (Prisma): id, name, slug, description, visibility, displayOrder.
- BadgeSkill junction table: composite PK (badgeId, skillId).
- CredlyOAuthToken model: stores OB3 access/refresh tokens for Credly auto-sync.
- 3 badges imported via Credly OB3: OPSWAT ICIP, Adobe Certified Professional, iTrain Asia CDSS.
- 1 skill ('Ai') linked to 1 badge (iTrain Asia CDSS).
- GET /api/skills — returns skills with badge counts and badge summaries.
- GET /api/badges — returns badge records with Credly embed data.
- GET /api/expertise/tree — endpoint exists, returns empty array [] (no nodes seeded).
- GET /api/experience — returns 46 training classes across 21 distinct topic categories.
- /skills page (test/agent-kit branch): basic grid layout, search, sort, pagination — no tree, no gamification.
- Live site renders /skills and /badges pages from deployed test/agent-kit build.

2.2 What Is Missing

- No visual skill tree — /expertise/tree returns empty, no tree rendering component exists.
- No domain-to-topic mapping layer connecting the 21 topic categories to 5 logical domains.
- No gamification layer — no XP calculation, no levels, no achievements.
- No panel interaction on skill nodes — clicking a skill does not show related training classes or badges.
- No pan/zoom controls on any tree view.
- Skills wallet does not show domain summary cards or domain-filtered class counts.
- Badges grid does not surface Credly verification links inline.
- No tabbed interface unifying the Skills Wallet and Skill Tree views.
- ExpertiseNode Prisma model (specified in CR-02) not yet implemented in either branch.

2.3 Gap Summary

Component	CR-02 Specified	Current State	CR-05 Resolution
Skill Tree	Specified (FR-53–FR-57)	Empty endpoint, no UI	Full radial SVG tree delivered
Domain Mapping	Implied	Not implemented	21 topics → 5 domains, hardcoded mapping
Tree Interaction	Slide panel (FR-56)	None	Click node → detail panel with classes + badges
Pan / Zoom	FR-53 zoom/pan	None	Mouse drag + scroll zoom + reset button
Gamification	Not in any prior CR	None	XP formula, 5 levels, 5 achievements
Skills Wallet	Partially in CR-01	Basic grid only	Domain chips, badge grid, skill cards, search

Component	CR-02 Specified	Current State	CR-05 Resolution
ExpertiseNode DB	Specified in CR-02	Not implemented	Deferred — tree built from classes API instead

3. ALIGNMENT WITH EXISTING SRS & CRs

CR-05 is fully aligned with the architectural and content-management philosophy established in SRS v1.0 and extended across CR-01 through CR-04. All new functionality is delivered as a single Next.js page component consuming existing API routes. No changes to authentication, user roles, deployment model, or existing database tables are required.

DOES NOT introduce new user roles	✓
DOES NOT change authentication model (JWT/NextAuth)	✓
DOES NOT alter deployment architecture (Vercel)	✓
DOES NOT modify existing Prisma schema (uses existing Badge, Skill, Class models)	✓
EXTENDS /skills page with tree, gamification, and domain intelligence	✓
CONSUMES only existing public API endpoints (/api/skills, /api/badges, /api/experience)	✓
DELIVERS production-ready implementation code (skills-page.tsx) alongside this CR	✓
DEFERS ExpertiseNode DB implementation (CR-02) — tree built from live class data instead	✓

4. SCOPE OF CHANGE

4.1 In Scope

- Tabbed /skills page: Skills Wallet tab and Skill Tree tab in a single unified page.
- Skills Wallet: domain summary chips (5 domains, colour-coded), verified badge grid with Credly verification links, skill area cards with badge thumbnails and training class counts.
- Radial SVG skill tree: Root → Domain → Topic node hierarchy derived from /api/experience topic categories.
- Domain-to-topic mapping layer: 21 topic categories mapped to 5 logical domains (Management & Leadership, Technology & Engineering, Cybersecurity, Data & AI, Productivity & Collaboration).
- Click-to-inspect node panel: shows node description, class count, badge count, supporting badge thumbnails, list of training classes delivered under that topic.
- Pan-and-zoom controls: mouse drag to pan, scroll wheel to zoom, +/– buttons, reset to default view.
- Gamification XP engine: XP = (badge count × 100) + (class count × 10), computed client-side.
- Level progression system: 5 levels (Trainee, Practitioner, Specialist, Expert, Master Trainer) with animated XP progress bar.
- Achievement system: 5 achievements (First Badge, Triple Threat, Decade Trainer, Multi-Domain, Expert Level) with locked/unlocked visual state.
- Hero header: stat chips (badges, skills, classes, domains), achievement display, level indicator.
- Responsive design: adapts to mobile, tablet, and desktop. Tree view accessible on all screen sizes.
- No new npm dependencies: tree implemented in pure React + SVG. No D3, no external graph libraries.

4.2 Out of Scope (Future Phase)

- ExpertiseNode database model and /api/admin/expertise-nodes CRUD (specified in CR-02, deferred).
- Admin-controlled tree node management (add/edit/move nodes via backoffice).
- Animated edge transitions (currently static SVG lines).
- Mobile touch gesture support (pinch-to-zoom on touchscreen).
- Public shareable skill card per individual skill (individual /skills/{slug} page redesign).
- Gamification persistence across sessions (XP is computed live, not stored in DB).
- Leaderboard or comparative gamification (single-user portal only).

5. MODULE A — SKILLS WALLET (VERIFIED CREDENTIALS GRID)

The Skills Wallet is the primary credential showcase tab. It presents all Credly-verified badges, domain summary statistics, and skill areas in a structured, searchable grid. It serves as the client-facing proof of Sulfri's verified competencies.

5.1 Functional Requirements

FR ID	Requirement
FR-179	The /skills page must display two tabs: 'Skills Wallet' (default) and 'Skill Tree'. The active tab must be visually distinguished with an amber highlight.
FR-180	The Skills Wallet tab must display a domain summary row showing up to 5 domain chips. Each chip must show: domain icon, domain name, and class count for that domain. Chips must be clickable to filter the skill view by domain.
FR-181	Domain chips must use the following colour coding: Management & Leadership (#f59e0b), Technology & Engineering (#3b82f6), Cybersecurity (#ef4444), Data & AI (#8b5cf6), Productivity & Collaboration (#10b981).
FR-182	The Skills Wallet must display a Verified Badges section showing all PUBLIC badges retrieved from GET /api/badges. Each badge card must show: badge image (fallbackImageUrl), title, issuer name, and issue date.
FR-183	Each badge card must include an inline 'Verify ■' link that opens the Credly badge page (https://www.credly.com/badges/{credlyBadgeId}) in a new tab. The link must only appear on hover to avoid visual clutter.
FR-184	If a badge has no fallbackImageUrl, a placeholder icon must be shown.
FR-185	The Skills Wallet must display a Skill Areas section showing all PUBLIC skills from GET /api/skills?withBadges=true. Each skill card must show: skill name, description (if present), badge count, matching class count, and badge thumbnail images.
FR-186	Each skill card must include a 'View training classes →' link that navigates to the homepage with a pre-applied topic filter: /?topic={skillName}.
FR-187	Skill cards must display a domain icon and domain colour derived from the domain mapping table (Section 8.1).
FR-188	A search input must filter the Skill Areas section in real time by skill name.
FR-189	The Skills Wallet must degrade gracefully if no badges or skills exist: the respective sections must show a friendly empty state message, not an error.
FR-190	All badge images must use lazy loading (loading='lazy' attribute) to improve page performance.
FR-191	Badge cards must animate on hover: border colour transitions to amber, slight elevation shadow.
FR-192	Domain filter selection must be visually reflected: selected domain chip must show a stronger border, background, and glow. Clicking the same chip again deselects it (toggle).
FR-193	The Skills Wallet must be fully responsive: single column on mobile, 2-column grid on tablet, 3-column grid on desktop.

6. MODULE B — INTERACTIVE RADIAL SKILL TREE

The Interactive Radial Skill Tree is a pure React/SVG visualisation of Sulfri's complete training expertise, derived dynamically from the live class database. It provides an engaging, explorable map of domains, topics, and credentials — structured as a radial tree with the trainer at the centre.

6.1 Tree Architecture & Domain Mapping

The tree has 3 levels: Root (trainer) → Domain (5 nodes) → Topic (21 nodes). Topic nodes are derived from the distinct topicCategory values in the classes table. The domain mapping is a client-side constant as defined in the implementation reference (Section 8.1). No database changes are required for this approach.

Level	Node Type	Source	Count	Colour
0 (Centre)	Root	Static — trainer name	1	Amber #f59e0b
1	Domain	Derived from domain mapping constant	5	Per domain colour
2	Topic	Distinct topicCategory from /api/experience	21	Inherits domain colour

6.2 Functional Requirements

FR ID	Requirement
FR-194	The Skill Tree tab must render a radial SVG tree using a computed radial layout algorithm. The tree must be centred in the available viewport with responsive sizing via ResizeObserver.
FR-195	The tree must use a 3-level radial layout: Level 0 (root) at centre (radius 0), Level 1 (domain nodes) at radius 140px, Level 2 (topic nodes) at radius 280px. Radii are scaled by the zoom factor.
FR-196	Node sizes must differ by type: Root = 38px radius, Domain = 26px radius, Topic = 18px radius.
FR-197	Each node must display: filled circle (colour = domain colour, transparency when unselected), domain-coloured border, node label below, class count number inside the circle.
FR-198	The root node must display a star emoji (■) at its centre and a radial glow effect using an SVG radialGradient.
FR-199	Nodes that have one or more supporting badges must display a small amber dot indicator at the top-right of the node circle.
FR-200	Edges (lines connecting parent to child nodes) must be drawn in the child node's domain colour, with 25% opacity by default. Edges connected to a selected node must increase to 80% opacity and 2px stroke width.
FR-201	Topic-level edges must use a dashed stroke pattern (strokeDasharray='4 4') to visually distinguish them from domain-level edges.
FR-202	Clicking a node must open a detail panel. Clicking the same node again or pressing the × button must close the panel. Only one node may be selected at a time.
FR-203	The detail panel must display: node type label, node name, description, class count stat card, badge count stat card (if > 0), list of supporting badge thumbnails with issuer names, and a scrollable list of training classes delivered under that topic (title, client, date, mode).

FR ID	Requirement
FR-204	The detail panel class list must be limited to 10 visible entries with a '+N more' indicator if the actual count exceeds 10.
FR-205	Mouse drag on the SVG canvas must pan the tree (translate transform). The cursor must change to 'grabbing' during a drag operation.
FR-206	Scroll wheel on the SVG canvas must zoom the tree (scale transform). Zoom range must be constrained between 0.4x (minimum) and 2.0x (maximum).
FR-207	Three control buttons must be visible in the top-left corner of the tree canvas: Zoom In (+), Zoom Out (-), Reset View (■). Reset View must restore zoom to 1.0 and pan offset to (0, 0).
FR-208	A domain legend must be visible in the bottom-left corner of the tree canvas showing each domain name with its colour dot and icon.
FR-209	When no node is selected, the right panel must show: a 'Click any node to explore' prompt, drag/zoom help text, and a domain quick-stats list showing class count per domain.
FR-210	The tree must use an SVG filter (feGaussianBlur glow) applied to selected and hovered nodes to create a visual glow effect.
FR-211	The tree must degrade gracefully if no classes are returned from /api/experience: an empty state message must be shown. The root node and header section must still render.

7. MODULE C — GAMIFICATION ENGINE

The Gamification Engine transforms the Skills Wallet from a static credential list into a dynamic, progress-driven experience. Visitors instantly understand the depth and breadth of Sulfri's expertise through a familiar level-and-achievement system that communicates authority without requiring them to read every credential individually.

7.1 XP & Level System

XP (Experience Points) is calculated entirely client-side from data returned by existing API endpoints. No XP data is stored in the database.

XP Formula: XP = (Number of Badges × 100) + (Number of Classes × 10)

Level	Title	Minimum XP	Colour	Unlock Condition (current data)
1	Trainee	0 XP	#6b7280 (Grey)	Default starting level
2	Practitioner	200 XP	#10b981 (Green)	~2 badges OR 20 classes
3	Specialist	500 XP	#3b82f6 (Blue)	~5 badges OR 50 classes
4	Expert	1,000 XP	#8b5cf6 (Purple)	~10 badges OR 100 classes
5	Master Trainer	2,000 XP	#f59e0b (Amber)	~20 badges OR 200 classes

7.2 Achievement System

ID	Icon	Label	Description	Unlock Condition
first_badge	■	First Badge	Earned your first credential	badges >= 1
triple	■■	Triple Threat	Earned 3 or more badges	badges >= 3
decade	■	Decade Trainer	Delivered 10 or more classes	classes >= 10
multi_domain	■	Multi-Domain	Active in 3 or more domains	active domains >= 3
expert_level	■	Expert Level	Delivered 50 or more classes	classes >= 50

7.3 Functional Requirements

FR ID	Requirement
FR-212	The hero header of the /skills page must display the current XP total, current level title, and level number. These values must be computed client-side from live API data on every page load.
FR-213	An XP progress bar must be displayed below the level label. The bar must fill proportionally between the current level's minimum XP and the next level's minimum XP.

FR ID	Requirement
FR-214	The XP progress bar must use a gradient fill: left colour = current level colour, right colour = next level colour. The bar must include a CSS glow (box-shadow) in the current level colour at 50% opacity.
FR-215	When the user has reached the maximum level (Master Trainer), the progress bar must show 100% fill. The next level label must not be shown.
FR-216	Below the progress bar, the system must display how many XP remain until the next level, in the format: '{N} XP to {NextLevelTitle}'.
FR-217	The hero header must display all 5 achievement icons in a row. Unlocked achievements must be displayed with full opacity, amber border, and amber glow. Locked achievements must be displayed at 30% opacity with a greyscale filter.
FR-218	Each achievement icon must include a tooltip (HTML title attribute) showing: achievement label and description text.
FR-219	Achievement unlock status must be computed client-side from: badge count (from /api/badges), class count (from /api/experience), and active domain count (derived from class topicCategory values).
FR-220	The hero header must display 4 stat chips: Badges (from /api/badges count), Skills (from /api/skills count), Classes (from /api/experience total), Domains (computed active domain count). Each chip must be colour-coded to its domain.
FR-221	Stat chips must use colour-coded borders and backgrounds: Badges = amber, Skills = purple, Classes = blue, Domains = green.
FR-222	The current level title must be displayed in the level's designated colour (see Level Progression Table). It must not use a fixed colour.
FR-223	All gamification values (XP, level, achievements, stat chips) must update automatically when the underlying API data changes — no manual refresh required.
FR-224	Gamification data must NOT be stored in the database. All computation must be client-side. This decision may be revisited in a future CR if persistence across sessions is required.

8. IMPLEMENTATION REFERENCE

8.1 Domain-to-Topic Mapping Table

The following mapping is implemented as a client-side constant (DOMAIN_MAP) in the page component. It translates raw topicCategory strings from the classes database into 5 logical domains. This mapping must be kept in sync with any new topic categories added to the classes table.

Domain	Icon	Colour	Topic Categories Mapped
Management & Leadership	■	#f59e0b	leadership, Management, Agile / Scrum, Project Management, Process Improvement, Career Development, Professional Skills
Technology & Engineering	■■	#3b82f6	Programming, Cloud Computing, Emerging Technology, Digital Transformation, Automation / RPA
Cybersecurity	■	#ef4444	Cybersecurity
Data & AI	■	#8b5cf6	Data Analytics & Visualization, Data Analytics & AI, Data Management, AI / Automation, AI / Prompt Engineering, AI Fundamentals
Productivity & Collaboration	■	#10b981	Microsoft Office Productivity, Productivity / Collaboration

8.2 File Placement

File	Action	Notes
src/app/skills/page.tsx	Replace / create	Main implementation — delivered as skills-page.tsx
prisma/schema.prisma	No change required	Uses existing Badge, Skill, BadgeSkill, Class models
src/app/api/skills/route.ts	No change required	Existing endpoint consumed as-is
src/app/api/badges/route.ts	No change required	Existing endpoint consumed as-is
src/app/api/experience/route.ts	No change required	Existing endpoint consumed as-is
package.json	No change required	No new dependencies — pure React + SVG

9. DATABASE AMENDMENTS

CR-05 requires NO database schema changes. All three modules consume data exclusively from existing tables defined in SRS v1.0, CR-01, and the test/agent-kit branch deployment.

Table	Status	Used By
badges	Existing (CR-01)	Module A — Badge grid, Module B — Node badge indicators
skills	Existing (CR-01)	Module A — Skill area cards
badge_skills	Existing (CR-01)	Module A — Badge-to-skill association
classes	Existing (SRS v1.0)	Module B — Topic node generation, Module C — XP + achievement calculation
profile_settings	Existing (SRS v1.0)	Hero header — trainer name (root node label)
expertise_nodes	Not yet created (CR-02)	Not used in CR-05 — tree built from classes instead

Note: If ExpertiseNode-based tree management (CR-02) is implemented in a future sprint, the DOMAIN_MAP constant in skills/page.tsx must be replaced with a call to GET /api/expertise/tree. The tree rendering logic (computeRadialLayout, flattenTree, getEdges) is already structured to accept any TreeNode hierarchy and will not require changes to accommodate this migration.

10. API AMENDMENTS

CR-05 requires NO new API endpoints. All data is sourced from existing endpoints. The table below documents the exact calls made by the new page component.

Endpoint	Method(s)	Description
`GET /api/skills`	GET (public)	Called with ?withBadges=true&limit;=100. Returns skills with badge summaries. Used by Module A (Skill Areas) and Module B (node badge indicators).
`GET /api/badges`	GET (public)	Called with ?limit=100. Returns all PUBLIC badge records including credlyBadgeId, fallbackImageUrl, issuer, issueDate. Used by Module A (Badge Grid) and Module C (XP calculation).
`GET /api/experience`	GET (public)	Called with ?limit=200. Returns training classes with topicCategory, clientName, mode, startDatetime. Used by Module B (tree nodes) and Module C (XP + achievements).
`GET /api/expertise/tree`	GET (public)	Exists but currently returns []. Not actively consumed in CR-05 implementation. Reserved for future CR-02 ExpertiseNode migration.

11. NON-FUNCTIONAL REQUIREMENTS

11.1 Performance

- The /skills page must load and render the initial Skills Wallet tab in under 2 seconds on a standard 4G mobile connection.
- All three API calls (skills, badges, experience) must be made in parallel using Promise.all — never sequentially.
- SVG tree layout computation (computeRadialLayout) must complete in under 100ms for the current dataset (5 domains, 21 topics).
- Badge images must use lazy loading to avoid blocking the initial paint.
- Gamification values (XP, level, achievements) must be computed using React useMemo hooks to avoid unnecessary recalculation on every render.

11.2 Compatibility & Dependencies

- The implementation must use zero new npm dependencies. The SVG tree, pan/zoom, and gamification engine are pure React + SVG + Tailwind CSS.
- The component must be compatible with Next.js 14 App Router and the 'use client' directive.
- The component must use only APIs available in the existing shadcn/ui and lucide-react packages already installed in the project.
- ResizeObserver is used for SVG canvas sizing — supported in all modern browsers (Chrome 64+, Firefox 69+, Safari 13.1+).

11.3 Accessibility

- All achievement icons must include a title attribute with the achievement label and description.
- The tab switcher must be keyboard-navigable.
- Badge images must include descriptive alt text (badge title).
- The SVG tree is a visual element — a text-based summary of domain/topic counts must be available in the right panel as an accessible alternative.

11.4 Security

- No sensitive data is displayed on the /skills page. All data is sourced from public API endpoints.
- Badge verification links use target='_blank' with rel='noopener noreferrer' to prevent tab-napping.
- No user input is sent to any API from this page. The search filter is client-side only.

12. ACCEPTANCE CRITERIA

The feature set introduced in CR-05 is considered complete when ALL of the following criteria are verified on the live Vercel deployment:

Module A — Skills Wallet

1. The /skills page renders two tabs: 'Skills Wallet' (default active) and 'Skill Tree'.
2. Domain summary chips are visible, correctly colour-coded, and show accurate class counts.
3. All Credly badges are displayed with their image, title, issuer, and date.
4. The 'Verify ■' link on each badge opens the correct Credly page in a new tab.
5. Skill area cards show badge count, class count, badge thumbnails, and domain icon.
6. The search input filters the Skill Areas section in real time.
7. Domain chip selection filters the view correctly. Clicking again deselects.
8. Empty states display gracefully if no badges or skills exist.

Module B — Skill Tree

1. The Skill Tree tab renders a radial SVG tree with root, domain, and topic nodes.
2. All 5 domain nodes are visible with correct colours and class counts.
3. All 21 topic nodes are visible, correctly mapped to their parent domain.
4. Clicking a topic or domain node opens the detail panel on the right.
5. The detail panel shows: node name, class count, badge count (if any), badge thumbnails, and list of training classes.
6. The detail panel × button correctly closes the panel.
7. Mouse drag pans the tree. Scroll wheel zooms in/out within the 0.4x–2.0x range.
8. +/– and reset (■) control buttons function correctly.
9. The domain legend is visible in the bottom-left corner.
10. Selected and hovered nodes display the glow filter effect.
11. Badge indicator dot is visible on nodes with linked badges.

Module C — Gamification

1. The hero header displays the correct XP total, level number, and level title.
2. The XP progress bar fills proportionally with a gradient glow matching the current level colour.
3. 'N XP to [NextLevel]' is shown when not at maximum level.
4. All 5 achievement icons are visible. Unlocked achievements show amber glow; locked achievements are greyed.
5. All 4 stat chips (Badges, Skills, Classes, Domains) show correct values.
6. Current level title is rendered in the correct level colour.
7. No database writes occur when loading this page.

13. IMPACT ON DEPLOYMENT

Area	Change	Impact
Infrastructure	None	No additional services required
Hosting (Vercel)	None	No model change, no new env variables
Database	None	No migrations required
Prisma Schema	None	No schema changes
npm Dependencies	None	Zero new packages
Files Changed	1 file: src/app/skills/page.tsx	Replace existing skills page with delivered component
Build Time	No significant change	SVG layout is runtime computation, not build-time
API Routes	None	No new or modified routes
Environment Vars	None	No new environment variables

Deployment instruction: Copy skills-page.tsx to src/app/skills/page.tsx in the repository. Commit and push to the deployment branch. Vercel will auto-deploy. No database migration, no build configuration changes, and no environment variable updates are needed.

14. VERSION CONTROL & CR LOG

Upon approval of CR-05, the main SRS version must be updated to v1.5. This CR must be merged into Sections 5 (Functional), 8 (Non-Functional), and 10 (Acceptance Criteria). The implementation file (skills-page.tsx) must be committed with the message: 'CR-05: Skills Wallet + Skill Tree + Gamification'.

CR ID	Target Version	Summary
CR-01	v1.1	Credly Badge Wallet & Skills Wallet Integration
CR-02	v1.2	Certificate-to-Training Expertise Linking + Visual Skill Tree (concept)
CR-03	v1.3	Event Lead Capture & YAYASAN PENERAJU Course Registration Module
CR-04	v1.4	Client-Facing Showcase Enhancement, Downloadable Profile, Automation & BI Module
CR-05	v1.5	Skills Wallet, Interactive Radial Skill Tree & Gamification Engine

15. STRATEGIC VALUE STATEMENT

CR-05 transforms the /skills page from a static list of credentials into the most compelling section of the Sulfri Trainer Portal — a living, interactive map of verified expertise that communicates depth, breadth, and credibility at a glance. The strategic outcomes are:

Instant Credibility Signal	A client who lands on the Skills page immediately sees a structured map of 5 domains, 21 topic areas, 46+ classes, and Credly-verified badges — all in a single, explorable view. This replaces the need to read a CV or sift through a list of courses.
Expertise Depth Communication	The radial tree makes it visually obvious how Sulfri's expertise is structured — not a flat list, but a coherent knowledge architecture. Clients in government procurement and corporate L&D; can immediately identify relevant domains without asking questions.
Gamification as Professionalism Signal	The XP level system reframes credentials and training history as a progression story. Showing 'Master Trainer' status with a visible progress bar communicates continuous professional development in a format that resonates with modern digital audiences.
Zero Maintenance Overhead	The tree and gamification values are derived entirely from existing database records. Every new class added in admin, every new Credly badge synced, automatically updates the skill tree, the XP bar, and the achievement unlocks — no admin action required.
Zero Infrastructure Cost	The entire module requires no new dependencies, no new database tables, no new API routes, and no new hosting configuration. It is a single-file deployment into the existing Next.js application.

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